

Last time Basic Axioms

Addition (P1) $(a+b)+c = a+(b+c)$

(P2) $\exists a$ s.t. $\forall a$

$$0+a = a+0 = a$$

(P3) $\forall a \exists "-a"$ s.t.

$$a+(-a) = 0 = (-a)+a$$

(P4) $\forall a, b \quad a+b = b+a$

Multiplication (P5) $\forall a, b, c$

$$(ab)c = a(bc)$$

(P6) $\exists 1$ s.t. $0 \neq 1$

$$\forall a, 1 \cdot a = a \cdot 1 = a$$

(P7) $\forall a \neq 0 \exists "a^{-1}"$ s.t.

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

(P8) $\forall a, b \quad a \cdot b = b \cdot a$

* If a system satisfies (P1) - (P4)
then it is an abelian group.

Combination (P9)

$\forall a, b, c$

$$a \cdot (b+c) = ab+ac$$

* If a system satisfies (P1)-(P9), it is a field

* If a system satisfies (P1)-(P9) except (P7), it is a Commutative ring.

Except (P7), (P8), it is a ring.

Prop. Assume (P1)-(P8)

Any equations of the form $ax+b=c$ has a solution.
($a \neq 0$)

PF We will show this by explicitly finding a solution to this equation.

Idea move b to another side of " $=$ ".

By (P3) the add inverse of b exists ($-b$).

Add $(-b)$ to both sides of equ.

$$ax+b+(-b) = c+(-b)$$

This is equiv. to (by (P_3))

$$ax + 0 = c - b$$

By (P_2) . now we have

$$ax = c - b$$

By (P_7) a^{-1} exists (since a is assumed to be not 0)

Multiply a^{-1} on both sides

$$a^{-1}ax = a^{-1}(c - b)$$

$$\text{By } (P_5) \quad (a^{-1}a)x = a^{-1}(c - b)$$

$$\text{By } (P_7) \quad 1 \cdot x = a^{-1}(c - b)$$

$$\text{By } (P_6) \quad x = a^{-1}(c - b) \quad \leftarrow \text{this is the solution of the equ.}$$



Prop For any a , $a \cdot 0 = 0$

Pf $0 + 0 = 0$ by (P2) Additive Identity

Then plug in $0 + 0$ for 0

$$\begin{aligned} \underset{(P9)}{a \cdot 0} &= a \cdot (0 + 0) \\ &= a \cdot 0 + a \cdot 0 \end{aligned}$$

By (P3) $(-a)$ exists so add $(-a)$ to both sides

$$(-a) \cdot 0 + a \cdot 0 = (-a) \cdot 0 + a \cdot 0 + a \cdot 0$$

$$\text{By (P3), (P1), } 0 = 0 + a \cdot 0 \stackrel{(P2)}{=} a \cdot 0$$



Proof $(-a) \cdot (-b) = ab$

Pf Strategy show that the additive inverse of $a \cdot b$ is $-(-a) \cdot (-b)$.

$$\begin{aligned} \text{Step 1. } ab + (-a)b &\stackrel{(P_1)}{=} (a + (-a))b \\ &\stackrel{(P_3)}{=} 0 \cdot b \\ &= 0 \rightarrow \text{by previous prop.} \end{aligned}$$

$$\begin{aligned} \text{Step 2 } &-(ab) + (-a)(-b) \\ \text{by step 1} &= (-a)b + (-a)(-b) \\ (P_1) &= (-a)(b + (-b)) \\ (P_3) &= (-a) \cdot 0 \\ &= 0 \rightarrow \text{by prev. prop.} \end{aligned}$$

$\Rightarrow$ Addl inv. of $(-ab)$ is $(-a)(-b) = ab$

Ex. Which of (P1)-(P9) are satisfied in:

① $\{\dots, -2, -1, 0, 1, 2, \dots\}$ integers $\mathbb{Z}$

Multiplication inv. DNE, but prop 2, 3 don't use this property so they are still true. But (P1) does use it, so it doesn't necessarily hold.

② $\{1, \dots, 12\} = \mathbb{Z}/12\mathbb{Z}$

Multi. inv. DNE; so prop 1 is false, but prop 2, 3 hold.

③ $\mathbb{Z}/11\mathbb{Z} = \{1, 2, \dots, 11\}$

Multi. inv. exist $\Rightarrow$ 11 is a prime #

④ $\{a/b : a, b \in \mathbb{Z}, b \neq 0\} = \mathbb{Q}$

⑤ $S = \{c(a, b, c) : a, b, c \in \mathbb{R}\}$

$+: S \times S \rightarrow S$

$(a_1, b_1, c_1), (a_2, b_2, c_2) \mapsto (a_1 + a_2, b_1 + b_2, c_1 + c_2)$

$\chi: S^2 \rightarrow S$  cross product.

$$(a_1, b_1, c_1), (a_2, b_2, c_2) \mapsto (b_1 c_2 - c_1 b_2, c_1 a_2 - a_1 c_2, a_1 b_2 - b_1 a_2)$$